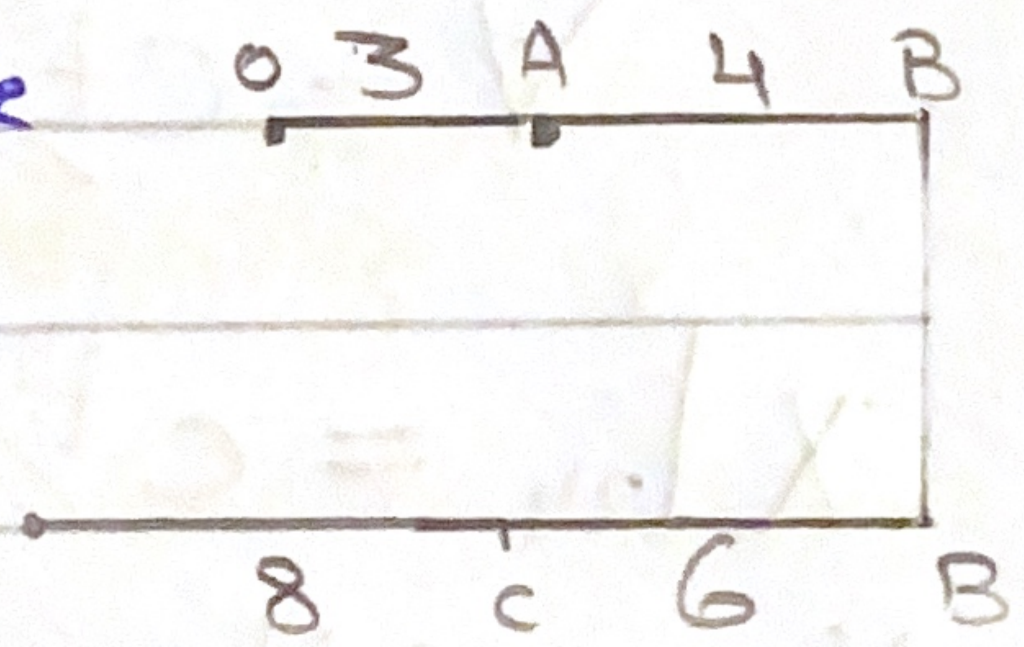
lecture 1

Kinematics of a particle:

→ Position (m)
→ Velocity (m/sec)
→ accelration (m/sec^2)

	Position	displacment	distance
A	+3	+3	+3
B	+7	+7	+7
C	+1	+1	+13



Q1. $x = t^3 - 9t^2 + 24t - 8$

ⓐ Determine when Velocity is zero.
ⓑ Position and total distance when the acceleration zero.

$V = 3t^2 - 18t + 24$ → ⓐ $0 = 3t^2 - 18t + 24$

$a = 6t - 18$ $\qquad t = 4 \quad t = 2$

ⓑ $0 = 6t - 18$

$t = 3$

$\text{distance} = |X_3 - X_2| + |X_2 - X_1|$

$= |-2| + |20|$

$= 22\,m$